

The Responsibility Gap in AI: Fairness Metrics and Privacy Tools as Sociotechnical Infrastructures

A Research Paper submitted to the Department of Engineering and Society

Presented to the Faculty of the School of Engineering and Applied Science
University of Virginia • Charlottesville, Virginia

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science, School of Engineering

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Spring 2026

On my honor as a University Student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments

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The rapid integration of artificial intelligence into critical social systems has created a significant sociotechnical problem known as the accountability gap in AI governance (Raji et al., 2020). Institutions increasingly rely on standardized mathematical metrics and procedural checklists to manage fairness and privacy. However, these tools frequently fail to prevent real-world harms to marginalized groups and can sometimes even obscure those harms from public view. For example, investigative audits of predictive policing and healthcare resource algorithms have consistently revealed stark racial disparities, despite these tools' reliance on ostensibly “objective” mathematical metrics (Angwin et al., 2016). Subsequent analyses reveal that algorithmic auditing often prioritizes technical compliance over meaningful social accountability, and thus exacerbating this gap (Selbst et al., 2019; Raji et al., 2020). This problem matters deeply because it creates a fundamental tension between the technical optimization of an algorithm and the lived social reality of the people subjected to its decisions. When harmful systems are allowed to operate under the guise of managed risk, they perpetuate systemic injustice while successfully shielding institutions from legal and public liability.

To analyze this disconnect, this research applies the theoretical framework of the ethnography of infrastructure developed by Star (1999). This framework displays that infrastructure is socially constructed, embeds specific values that shape behavior, and remains largely invisible until it breaks down. Conceptually, this paper treats fairness metrics, privacy definitions like differential privacy, and bureaucratic governance audits not as neutral tools, but as distinct sociotechnical infrastructures. The primary research question guiding this study asks how current technical and bureaucratic AI governance tools function as infrastructures, and why they frequently fail to achieve algorithmic justice. By relying on a qualitative analysis of academic literature, policy evaluations, and empirical studies regarding model accuracy, this

paper argues that current fairness and privacy mechanisms function as sociotechnical infrastructures designed to stabilize the appearance of responsible governance. By relying heavily on abstraction and standardization, these infrastructures prioritize institutional liability and procedural efficiency over the rectification of injustices.

To support this argument, the paper is structured around four key sub-arguments. First, it demonstrates how mathematical fairness metrics create an abstraction trap that standardizes complex grievances into reductive variables. Second, it examines how privacy technologies act as a paradox that protects data secrecy while degrading system utility for vulnerable populations. Third, it analyzes how bureaucratic governance frameworks, such as bias audits, function as symbolic infrastructures that produce null compliance rather than actual accountability. Finally, it explores the power dynamics inherent in defining these metrics, arguing that the technical definitions of fairness serve to maintain existing corporate power structures rather than dismantle them.

The use of mathematical fairness metrics commonly used by institutions is insufficient for closing the responsibility gap, and should not be a primary piece of evidence to validate governance. Mathematical fairness metrics are statistical formulas that engineers use to constrain a machine learning model's outputs so that they balance out across different demographic groups. These metrics serve as a vital part of the sociotechnical infrastructure. According to Star (1999), a key characteristic of infrastructure is its use of master narratives, which standardize a particular activity into a standardized form. In the case of AI governance, fairness metrics serve this exact purpose, standardizing the complex and messy concerns of marginalized communities into a set of variables that can be optimized by an engineering team. This is a very appealing thing to institutions because it allows the concept of fairness to be contained within the strict technical

process. However, it also creates what Selbst et al. (2019) describe as an “abstraction trap.” By strictly defining fairness as a property of the mathematical output of a model, these metrics remove the social context that is necessary for understanding the underlying causes of inequity. This simplifies the problem, but it serves an administrative purpose rather than a moral one, as it allows institutions to say that they are addressing issues of bias without ever having to actually deal with the underlying realities of the issue.

Barocas et al. (2023) offer the technical defense of this approach by defining fairness as a constraint that can be placed on a model during training. They see these metrics as crucial translation tools because, without them, fairness would be a mere philosophical ideal that is beyond the reach of the logic of an algorithm. This approach, therefore, essentially translates a moral problem into an engineering challenge. Although I agree that some form of measurement is required for regulation, the use of these metrics can paradoxically lead to the mathematical proxy being conflated with reality itself. When fairness is turned into a mathematical proof, the infrastructure becomes invisible in the way described by Star (1999). The assumptions encoded in the code, such as the assumption that fairness is achieved when error rates are equal for all groups, become invisible. They become the new standard of the system, successfully insulating developers from the complex social realities the algorithm is actually meant to regulate. The process of measuring fairness, therefore, paradoxically leads to the prevention of justice being achieved.

This standardization is undermined by the abstraction trap identified by Selbst et al. (2019). They show that technical metrics will always fail because they do not consider the social systems in which the algorithm operates. For example, an algorithm may be technically fair by balancing the prediction rate for two different demographics. But if one demographic is more

aggressively policed in the real world because of past bias, then the algorithm's equal predictions will simply perpetuate the systemic injustice. In this case, the metric is working beautifully as a piece of technical infrastructure because it is optimized and standardized. But it is completely failing as a means of justice. This is a huge mismatch in how we regulate AI. By optimizing for computable fairness, institutions are actually optimizing for their own risk management and efficiency, not for justice. The metric is simply a shield that allows the system to be safe because it balances out, making the experience of injustice invisible to the institution. The infrastructure succeeds only in that it preserves the status quo of the institution, rather than challenging the cycle of harm that it purports to measure.

Strict adherence to mathematical privacy definitions, specifically differential privacy, frequently undermines the utility of AI systems for vulnerable populations and fails to align with social expectations of safety. Like fairness metrics, differential privacy is a technical fix that enables organizations to use data while maintaining that they are protecting individual rights. Star (1999) points out that infrastructure is often nested inside other infrastructure. Differential privacy is nested directly into data streams to provide mathematical guarantees. It does this by adding noise to the data stream to ensure that the result cannot possibly be used to determine whether the data of a particular individual was used. This provides a strong mathematical guarantee of secrecy, but it creates a paradox in which the infrastructure meant to protect the vulnerable is actually degrading the utility of the systems that the vulnerable rely on. This nesting of values creates a situation in which the technical notion of safety is actually working against the needs of the user.

Abadi et al. (2016) describe the huge technical success of differential privacy in deep learning. They show that this approach can successfully protect against membership inference

attacks without completely breaking the model's ability to learn. From a technical point of view, Abadi et al. tell a success story about how privacy is achieved by adding statistical noise to the model. However, I argue that Abadi et al.'s technical validation is insufficient for sociotechnical contexts because it completely ignores the disproportionate cost of this noise on marginalized groups. Bagdasaryan et al. (2019) offer a compelling argument against this success story. Their study shows that the noise added to protect privacy disproportionately hurts the accuracy of the model for underrepresented groups. Since minorities, to begin with, have less data in the training set, the privacy-preserving noise drowns out the signal of their data. This makes the infrastructure of protection an instrument of exclusion, where the price of privacy is paid in the currency of model accuracy.

This leads to a chilling sociotechnical reality in which the price of privacy is not equally paid. To ensure the privacy of the minority user, the system has to degrade the algorithm considerably to recognize or serve the minority user. Bagdasaryan et al. (2019) show that the infrastructure of differential privacy is not neutral. Rather, it imposes a tax on accuracy, which is paid largely by those on the margins. This is consistent with the observation by Star (1999) that infrastructure for one person can be a barrier for another. For the majority, the system functions with slightly lower accuracy, but for the minority, the system can become unusable. Moreover, the use of differential privacy reveals a philosophical weakness in how engineers currently define privacy. The technical definition of secrecy is at odds with the notion of contextual integrity introduced by Nissenbaum (2009). I agree with Nissenbaum's contention that privacy must be defined by appropriate information flow rather than mere secrecy, as this framework exposes the failure of technical tools that protect data but harm people. A system that preserves identity using differential privacy but fails to provide a loan because the noise damaged the accuracy of the

model about a particular demographic has violated the contextual integrity of the user. The system maintained the secrecy of the data but did not respect the social context of the user, thus proving that technical solutions are often at odds with social expectations. The system's prioritization of secrecy over social context ignores the complex social flows that define privacy in a connected world.

The infrastructure of AI governance, while generally designed to ensure accountability through audits and transparency notices, frequently functions as a primarily symbolic system that prioritizes the appearance of compliance over the actual solutions of harms. I argue that this mostly bureaucratic focus actively stabilizes institutions by creating an observable output of ethics that can be filed and approved, regardless of whether the underlying algorithmic bias has been mitigated. This aligns with the observation by Star (1999) that infrastructure is learned as part of membership within a community. In the corporate world, the learned behavior is compliance. Organizations learn to perform the rituals of accountability as a condition of their membership in the regulated market. Consequently, the effectiveness of the tool matters less than the certificate of the audit. The bureaucracy serves to legitimate the institution by producing a paper trail of ethics, irrespective of the actual impact on the vulnerable.

A prime example of this performative infrastructure can be seen in the recent implementation of New York City Local Law 144. This legislation mandates that employers using automated employment decision tools to hire or promote candidates must first subject those algorithms to an independent bias audit. Furthermore, the law requires employers to publish a public transparency notice on their website. In theory, a transparency notice is a mechanism designed to empower job seekers by informing them that an algorithm is evaluating their application, detailing the data being collected, and providing a formal channel to request

alternative evaluation methods. The intended implication of this law is to create a legally binding system of informed consent and fairness. However, the practical application of this law reveals the deep flaws of relying on bureaucratic solutions. Wright et al. (2024) analyze NYC Local Law 144 and characterize it as producing null compliance because the resulting transparency notices are often buried in dense corporate websites, written in inaccessible legal jargon, and practically impossible for workers to act upon.

This finding is strongly supported by Metcalf et al. (2019), who argue that the institutionalization of ethics in the technology sector often converts broad moral goals into narrow corporate procedures that fit existing business models. I agree with the assessment by Metcalf et al. that this bureaucratic translation neuters the original intent of the regulation. By turning justice into a checklist, the infrastructure serves the institution's need for liability protection rather than the public's need for equity. Greene, Hoffman, and Stark (2019) expand on this by describing how high-level ethical principles are often deployed as a form of moral branding. They argue that vague commitments to fairness or transparency allow companies to avoid specific, binding regulation while projecting an image of responsibility. This branding acts as a preemptive defense, creating a soft infrastructure of voluntary codes that distracts from the need for hard legal constraints.

This performative aspect creates a significant and often overlooked burden on the individuals subjected to these systems. Ananny and Crawford (2018) describe this as a transparency fallacy, where the mere disclosure of information is conflated with accountability. By relying on public notices as the primary mechanism of governance, institutions effectively shift the labor of monitoring from the regulator to the citizen. This presumes a level of technical literacy and available time that most affected individuals simply do not possess. Consequently,

the infrastructure of transparency is designed for a privileged user who often does not exist in the context of marginalized communities. Sun et al. (2025) identify this as a structural meta-problem in algorithmic fairness. They argue that when fairness tools are adopted symbolically, they create a closed loop where the institution satisfies its own internal metrics without ever providing a mechanism for external contestability. The transparency notice becomes a dead end rather than a starting point for recourse, effectively silencing those who attempt to navigate the bureaucracy to challenge an algorithmic decision. This reinforces Star's (1999) concept that infrastructure is fundamentally relational. To the corporation, the transparency notice is a completed task; to the worker, it is an illegible barrier that obscures the decision-making process rather than revealing it.

On the other hand, proponents of standardization like Schwartz et al. (2022), who developed the NIST AI Risk Management Framework, argue that rigorous risk mapping is the most effective path to safety. They contend that by standardizing how we identify and measure bias, institutions can systematically reduce harm. I disagree with the premise by Schwartz et al. that risk management alone is sufficient for justice. Their framework treats bias as a safety defect to be minimized, much like a software bug, rather than a fundamental contest over rights and power. Internal risk management often allows companies to grade their own homework. As Raji et al. (2020) point out, this leads to audit washing, a practice where a company subjects its AI to a superficial, low-rigor audit and then leverages the mere existence of that assessment as proof of algorithmic safety. Without external enforcement mechanisms that Schwartz et al. largely omit, these risk frameworks become performative routines. The reliance on voluntary standards risks establishing a governance model where the friction of regulation never actually slows down the

machinery of deployment, allowing harmful systems to operate indefinitely under the guise of managed risk.

The very process of defining and designing AI governance infrastructures is fundamentally a political act of power disguised as a neutral technical decision. When engineers choose which mathematical metrics to apply or which demographic groups to protect, they are encoding specific social hierarchies directly into the architecture of the system. Star (1999) emphasizes that infrastructure is fundamentally relational, noting that one person's infrastructure is frequently another person's barrier. In the context of algorithmic governance, the technical definitions of fairness determine exactly who is legible to the system and who is erased. This dynamic transforms the creation of accountability tools into a mechanism of control, where the power to define the terms of fairness is heavily concentrated within the institutions deploying the technology.

This power dynamic is highly visible in foundational technical approaches to bias. Dwork et al. (2012) introduce the concept of fairness through awareness, a framework that relies on a specific mathematical distance metric to define which individuals are similarly situated and therefore deserve equal treatment. However, I argue that the choice of how to define similarity is not an objective calculation but a highly subjective judgment that dictates the strict boundaries of legal and social protection. Defining this distance metric requires translating complex social dynamics into high-dimensional probabilistic spaces, a translation that engineers often lack the sociological expertise to perform accurately. By leaving these crucial definitions entirely to technical teams, the resulting infrastructure naturalizes the blind spots of its creators. It validates the perspective of the dominant group while presenting that perspective as an undeniable

mathematical truth, thereby erecting a barrier for anyone whose lived experience falls outside that calculated metric of similarity.

While Star's (1999) framework provides a crucial lens for diagnosing this responsibility gap, her theory of infrastructure struggles to fully capture the reality of corporate artificial intelligence. Star posits that infrastructure becomes invisible organically over time as it becomes a naturalized, functioning part of a community. However, the obfuscation of AI governance is rarely organic. Technology companies actively hide the metrics, weights, and training data of their models behind strict intellectual property laws and trade secrets. Furthermore, Star argues that infrastructure suddenly becomes visible upon breakdown. In the context of AI, systemic breakdowns in fairness are frequently dismissed by corporations as mere technical glitches or isolated anomalies. This active corporate framing prevents the infrastructure from becoming truly visible to the public. Therefore, applying Star's theory to AI reveals a critical limitation. Her framework assumes a somewhat passive process of normalization and underestimates the role of intentional, top-down corporate power in deliberately manufacturing the invisibility of proprietary digital systems.

This deliberate centralization of definitional power is explicitly critiqued by Kak and Myers West (2023) from the AI Now Institute. They argue that current audit-centered approaches to AI governance consistently centralize power in the hands of the technology companies creating the tools, effectively allowing these corporations to grade their own homework. I agree with their assessment that this self-regulatory environment is deeply flawed. When corporations control the infrastructure of accountability, they inevitably design it to validate their existing business models and limit their own legal liability rather than to empower marginalized communities. This dynamic reinforces the role of infrastructure in maintaining existing power

structures rather than dismantling them. Consequently, the responsibility gap is not merely an accidental failure of measurement. It is the result of a deliberate infrastructural design that successfully consolidates corporate power under the guise of procedural compliance.

By examining technical metrics, privacy technologies, governance laws, and the power dynamics of definition, this research demonstrates that responsible artificial intelligence is currently constructed as a rigid set of bureaucratic and technical infrastructures. These infrastructures are highly successful at making algorithmic systems manageable for institutions, but they consistently fail at making these systems just for people. The theoretical framework provided by Star (1999) binds these distinct examples together by highlighting the mechanism of invisibility. The standardization required by these infrastructures inherently erases the specific needs and contexts of marginalized groups, whether through the abstraction trap of mathematical fairness, the statistical noise of differential privacy, or the performative compliance of a bias audit.

The persistence of this responsibility gap matters because it proves we cannot resolve algorithmic bias solely through improved mathematics or stricter procedural checklists. This analysis reveals several critical implications for the future of technology development. Primarily, we have learned that because technical fixes inherently clash with social expectations, governance must fundamentally shift away from procedural compliance and toward outcome-based accountability. This implies that future regulatory solutions must incorporate social friction and context rather than simply optimizing for algorithmic efficiency. It also implies that true accountability necessitates transferring the power of definition away from self-regulating technology companies and toward external, community-centered bodies that

utilize frameworks like contextual integrity (Nissenbaum, 2009) to evaluate actual social impact rather than numerical parity.

While this research highlights the deep flaws in current governance models, it is limited by its heavy focus on regulatory frameworks centered around the United States, such as the NIST Risk Management Framework and New York City Local Law 144. Additionally, the critique acknowledges the reality that some form of standardized measurement remains necessary to regulate technology at a global scale. These limitations inspire crucial new research questions for the field of science, technology, and society. Future studies must ask how non-western or international regulatory frameworks, such as the European Union AI Act, manifest as sociotechnical infrastructure and whether they suffer from the same abstraction traps. Furthermore, researchers must ask how policymakers can successfully design anti-abstraction regulatory tools that mandate qualitative, community-led feedback without completely halting technological development. By exploring these questions, the field can begin to design accountability mechanisms that serve the public interest rather than merely protecting the institutions that deploy them.

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